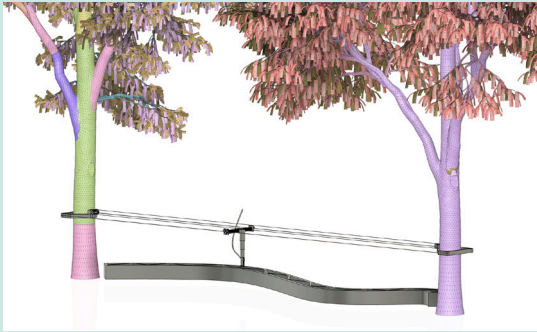
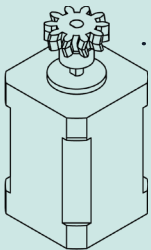




What do we mean by accessible robots?

Automation and assembly at architectural scale is being researched in many institutions and practices with various approaches, materials, and applications.

HLRE contributes to this field by exploring low-cost, site-agnostic strategies to implement emerging and programmable methods of building. We imagine this backyard robot assisting the home builder or DIY enthusiast, offering tangible access points to robotic construction methods that have been out of reach due to cost and methodology.



Accessible Robots

Hyper-Local Robotic Entanglements is a project led by FSD.TEMP that explores the design and implementation of a DIY backyard robot to build with natural materials gathered from local sources. HLRE explores how to encourage the tangibility of emerging technologies through public engagement and physical artifacts.

FSD.TEMP Team

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Connect with us!
Instagram: @funstuffdesign
<https://funstuffdesign.com>

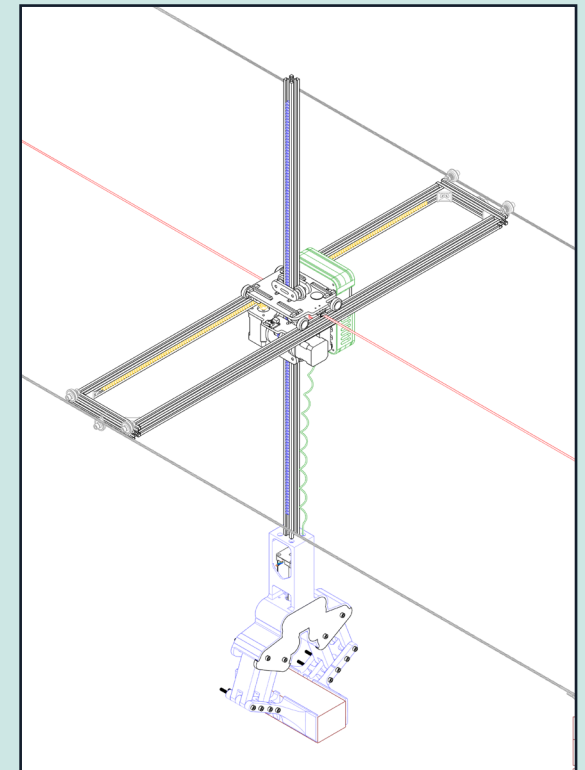
FSD.TEMP



GitHub
Repository

FSD.TEMP

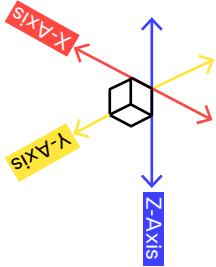
Backyard Bot: community- accessible building robot



Hyper-Local
Robotic Entanglements

2025-2026

Backyard Bot Plans



SOFTWARE

This machine runs on Klipper, an opensource firmware used for 3D printer control. Reference our online documentation for our custom setup.

3D PRINTED PARTS

- (Core)
 - 2 DoubleHelicalPinion
 - 12 DoubleHelicalRack
 - 1 CapstanDrum
 - 1 EncoderDrum
 - 1 CaseTOP
 - 1 CaseBOTTOM
- (Gripper)
 - 1 servo motor housing
 - 1 upper shaft with gear
 - 1 lower shaft with cover
 - 1 gripper housing upper
 - 1 gripper housing lower
 - 1 threaded rod carriage
 - 2 parallel gripper link 1
 - 2 parallel gripper link 2
 - 2 parallel gripper link 3
 - 2 parallel gripper finger center
 - 4 parallel gripper finger ends
 - 1 servo gear

HARDWARE

This machine is built from an assembly of Purchased Parts, LaserCut Parts (from any flat-stock sheet good), and 3D printed parts. Hardware used is all off the shelf assembly. Full models and plans are available in our online repository.

PURCHASED PARTS - Amazon, McMaster, Adafruit, Local Hardware

FASTENERS

(Core)	Size	Name/Length (mm)
1	EBB42 Gen 2	
1	CB2	
1	Manta M5P board	
1	1kg roll PLA Filament	
1	1/4" Acrylic 3x4 sheet	
1	3/16 Stainless Steel Wire Rope 7x19 strand	6
1	100cm x 2cm T-Extrusion	10
2	16cm x 2cm T-Extrusion	4
1	7 Strand 550lb Type III Paracord	6
6	m5x20 bolt	6
4	m5x45 bolt	4
10	m5x35 bolt	10
1	m5x35 bolt	10
21	m5x8 bolt	21
21	m5 T-nut	21
16	m5 Nut	16
4	m5 Nylock Nut	4
6	m5 Heat Set Insert	6
14	m5 Washer	14
8	m3x18 bolt	8
4	m3x15 bolt	4
28	m3x12 bolt	28
4	m3x10 bolt	4
28	m3 Nuts	28
44	m3 Washers	44
4	m2x15 bolt	4
4	m3 Plastic Stand-offs	4
4	m3 Plastic Screws	4
12	m4x60 shoulder bolt	12
4	m4x30 bolt	4
16	m4 washers	16
16	m4 nuts	16
1	m5x110 bolt	1
4	m2x8 flat head bolt	4
4	m5x25 bolt	4
4	m5x35 bolt	4
8	m5 nut	8
1	m5 Nylock nut	1

Bill of Materials

FULL DOWNLOAD
AT GITHUB
REPOSITORY



- (Core)
 - 1 1/4" Acrylic 3x4 sheet
 - 1 1kg roll PLA Filament
 - 1 Manta M5P board
 - 1 CB2
 - 1 EBB42 Gen 2
 - 3 100cm x 2cm T-Extrusion
 - 2 16cm x 2cm T-Extrusion
 - 1 7 Strand 550lb Type III Paracord
 - 6 3/16 Stainless Steel Wire Rope 7x19 strand
 - 8 3D printer roller (creality ender type)
 - 3 608 Bearing
 - 3 Micro Limit Switch (normally open)
 - 2 NEMA 17 Stepper Motor
 - 1 NEMA 17 Stepper Motor with gear box (14:1)
 - 7 1/4 ID, 1 1/4 OD cable roller
- (Gripper)
 - 1 NEMA 17 Stepper Motor
 - 1 Servo motor 5v
 - 1 Rotating Flange
 - 1 End Stop
- CUT PARTS

- (Core)
 - 1 PCBBackPlate
 - 1 PCBFrontPlate
 - 1 ZPerpTOP
 - 1 ZPerpBOTTOM
 - 2 XPerp
 - 1 ZStepperSupport
 - 4 YWheelVert
 - 4 YWheelVertLock
 - 2 YPerpRollerGuideTOP
 - 2 YPerpRollerGuideBOTTOM
 - 2 acrylic face plates
- (Gripper)
 - 1 servo gear
 - 4 parallel gripper finger ends
 - 2 parallel gripper finger center
 - 2 parallel gripper link 3
 - 2 parallel gripper link 2
 - 2 parallel gripper link 1
 - 1 threaded rod carriage
 - 1 gripper housing upper
 - 1 gripper housing lower
 - 1 lower shaft with cover
 - 1 upper shaft with gear
 - 1 servo motor housing